Name: Sri hari siddharth M
Roll No: 241201055
Year & Dept: 3 & MCT

Experiment 3: Design of Test Cases and a Maintenance Plan for a Sample Software Module

1. Aim

To design test cases for the login and registration module of the Library Management System and to prepare a maintenance plan for the module after deployment.

2. Module Description

The login module accepts a registered user ID and password, validates the input format, checks the credentials against the database using a hashed password comparison, locks the account after repeated failed attempts, and provides a password reset option through a registered email address. The registration module collects personal details such as name, roll number, department and email, checks for duplicate user IDs, enforces password rules and sends a confirmation email.

3. Testing Strategy

The testing approach combines black-box and white-box techniques. Black-box methods include equivalence partitioning, for example valid and invalid password classes, and boundary value analysis, for example minimum and maximum password lengths. White-box checks verify the logic of the failed-attempt counter and the lock timer. Security testing is included with input that imitates common attacks, such as SQL injection. Each test case has an identifier, a scenario, input data, expected result and status, so that results are traceable.

4. Test Case Design

TC ID	Test Scenario	Input	Expected Result	Status
TC01	Valid login	Correct ID and password	Dashboard opens	Pass
TC02	Wrong password	Correct ID, wrong password	Error message shown	Pass
TC03	Empty fields	Blank ID and password	Validation message shown	Pass
TC04	Account lock	Five consecutive wrong passwords	Account locked for 15 minutes	Pass
TC05	SQL injection attempt	Quote-based injection string in ID field	Input rejected, no access	Pass
TC06	Password reset	Registered email address	Reset link sent	Pass
TC07	Duplicate registration	Existing user ID	Duplicate error shown	Pass
TC08	Email format	Invalid email string	Format error shown	Pass
TC09	Boundary password length	Seven characters (below minimum of eight)	Rejected with rule message	Pass

TC ID	Test Scenario	Input	Expected Result	Status
TC10	Session timeout	No activity for 20 minutes	User logged out automatically	Pass

All ten test cases were executed on the staging server. Any failed case would have been logged as a defect with its steps to reproduce, severity and the build number, and then retested after the fix. Each case was marked Pass after confirming both the on-screen message and the database state.

5. Defect Log Format

Defect ID	Module	Severity	Steps to Reproduce	Status
D-001	Login	High	Enter wrong password six times; lock not applied	Closed after fix
D-002	Registration	Low	Error text overlaps input box on small screens	Closed after fix

6. Maintenance Plan

Maintenance Type	Activity	Frequency
Corrective	Fix defects reported by users or testers	As required
Adaptive	Adjust to new browser versions, PHP or database upgrades	Quarterly review
Perfective	Improve usability, for example a password strength meter	Per release cycle
Preventive	Security patching, code refactoring, dependency updates	Monthly

- The regression suite (TC01 to TC10) is run after every change to the module, before release.
- Change requests are logged, prioritised and tested in a staging environment before going live.
- Logs of failed logins are reviewed weekly to detect signs of brute-force attacks.
- Backups of the user table are taken nightly and restore procedures are tested every quarter.
- Ownership of the module is assigned to one developer, with a named backup developer.

7. Discussion

Maintenance cost typically exceeds development cost over the life of a system, so a plan that anticipates change is valuable. The security test cases and the weekly log review are particularly important for a login module, because it is the main entry point to the application. The defect log shows that usability issues, not only functional issues, are found during testing and should be included in the regression suite.

8. Result and Conclusion

Ten test cases covering functional, boundary and security aspects of the module were designed and executed, and a maintenance plan defining corrective, adaptive, perfective and preventive activities was prepared. Together they ensure the module remains reliable, secure and up to date throughout its operational life.